Motor Insurance Risk Analytics: Claim Frequency, Severity and Aggregate Loss Modelling

Portfolio experience from 2022–2024

Abstract

This report analyses anonymised motor insurance policy-year data from 2022 to 2024. Annual claim frequency increased from 0.302 to 0.312 claims per unit of exposure, while the incurred-loss ratio rose from 65.6% to 74.7%. A Negative Binomial model handled overdispersion better than a Poisson benchmark, although a 2024 holdout test underpredicted the number of claims by 3.4%. Claim severity was strongly-right skewed, with a mean of approximately €1,019 and a median of €644. The Lognormal distribution fit better overall than the Gamma distribution, but both underestimated the most extreme observations. A Generalized Pareto tail model produced substantially higher simulated VaR and TVaR than the Lognormal benchmark. These estimates should be treated cautiously: severity observations are policy-year average losses per claim count rather than individual claim transactions, and the highest-tail estimates are sensitive to simulation size.

1. Introduction

Motor insurance risk assessment requires insurers to understand both how frequently claims occur and the financial cost associated with those claims. These two components of risk are influenced by policyholder and vehicle characteristics, and their combined behaviour determines the potential losses experienced by an insurance portfolio. Relatively infrequent, high-cost claims can have a substantial influence on aggregate losses, making the accurate modelling of claim severity and its upper tail an important part of risk assessment.

This project investigates motor insurance portfolio experience using policy-year data covering 2022–2024. The analysis begins with data validation and cleaning, followed by exploratory data analysis and SQL-based portfolio summaries using PostgreSQL. Statistical modelling is then conducted in R to investigate claim frequency and positive incurred claim severity. Poisson and Negative Binomial regression models are considered for claim counts, while Gamma and Lognormal distributions are fitted to severity observations. Extreme Value Theory (EVT) is subsequently used to model high-severity observations that may not be adequately represented by standard parametric distributions.

The final stage combines the frequency and severity modelling components within a Monte Carlo simulation to estimate the portfolio-level aggregate loss distribution. Value at Risk (VaR) and Tail Value at Risk (TVaR) are calculated at selected confidence levels to investigate how extreme losses may affect the portfolio. The analysis also considers model diagnostics, out-of-

sample frequency validation and the sensitivity of aggregate risk estimates to the treatment of extreme severities.

The overall objective is to develop an integrated analytical workflow for understanding motor insurance risk, evaluate the suitability of alternative statistical models, and assess the importance of extreme claims in aggregate-loss estimation. The findings are interpreted considering the assumptions and limitations of the available data, particularly the use of policy-year average severities as a proxy for individual claim amounts.

2. Data and preparation

2.1 Dataset and analytical samples

The analysis uses the publicly available dataset *A detailed dataset of motor insurance policies with coverage-specific financial information*, published by Espinosa, Lledó and Atance (2026). The dataset contains anonymised motor insurance portfolio data covering 2022–2024, including policyholder and vehicle characteristics, exposure, premiums, claim counts and incurred losses. The original dataset was downloaded from Mendeley Data and subsequently cleaned and validated in R before being analysed using PostgreSQL and statistical modelling techniques.

The dataset contains 354,140 anonymised policy-year records covering motor insurance business in Spain from 2022 to 2024. It includes driver, vehicle and policy characteristics alongside exposure, premium, claim-count and incurred-loss measures (Espinosa, Lledó and Atance, 2026). Since the same insured may appear in multiple years, records represent policy-year observations rather than distinct policyholders.

After cleaning and validation, 354,091 positive-exposure observations were retained for frequency modelling. Forty-nine records had zero exposure and were excluded from this analysis. Of these, 48 had zero claims, premium and incurred loss, while one recorded a claim and €180.11 of incurred loss despite having zero exposure. The severity sample contained 42,263 records with positive claim counts and positive incurred losses. A further 8,543 observations had claims recorded but no incurred loss, so they were not included in the positive-severity sample.

Data cleaning included checking missing values, numeric ranges and consistency across the financial variables. SQL queries in PostgreSQL were used to summaries experience by year and policy type, while R was used for exploratory analysis, model fitting, diagnostics and simulation.

2.2 Annual portfolio overview

The volume of policy-year records rose considerably over the period, from 67,172 in 2022 to 168,133 in 2024. Reported incurred losses increased in each year. The following table gives the main annual totals; the frequency measure is claims divided by exposure, and the loss ratio is incurred losses divided by premium.

Year	Policy-year records	Claims	Premium (€)	Incurred losses (€)	Frequency	Loss ratio
2022	67,172	12,665	18,697,928	12,270,105	0.302	65.6%
2023	118,835	25,158	35,344,308	23,488,354	0.305	66.5%
2024	168,133	39,276	51,037,433	38,106,351	0.312	74.7%

Table 1. Annual portfolio experience, 2022–2024.

3. Exploratory analysis

Annual claim frequency rose gradually, from 0.302 claims per unit of exposure in 2022 to 0.312 in 2024. The increase is modest, but it suggests that claim frequency was not entirely stable across the period.

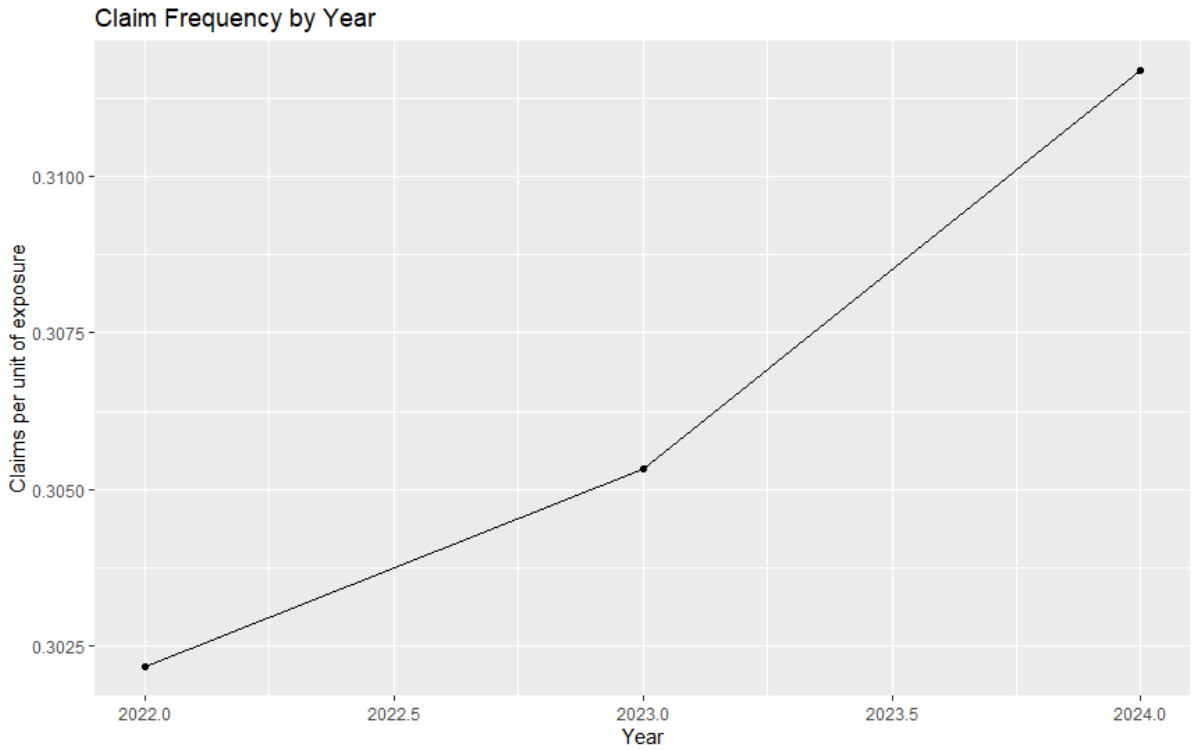


Figure 1. Annual claim frequency, 2022–2024.

The incurred-loss ratio increased more sharply, from 65.6% in 2022 to 74.7% in 2024. This means incurred losses grew relative to premium income. The ratio alone does not identify the cause: changes in claim frequency, claim costs, premium levels and portfolio composition may all contribute.

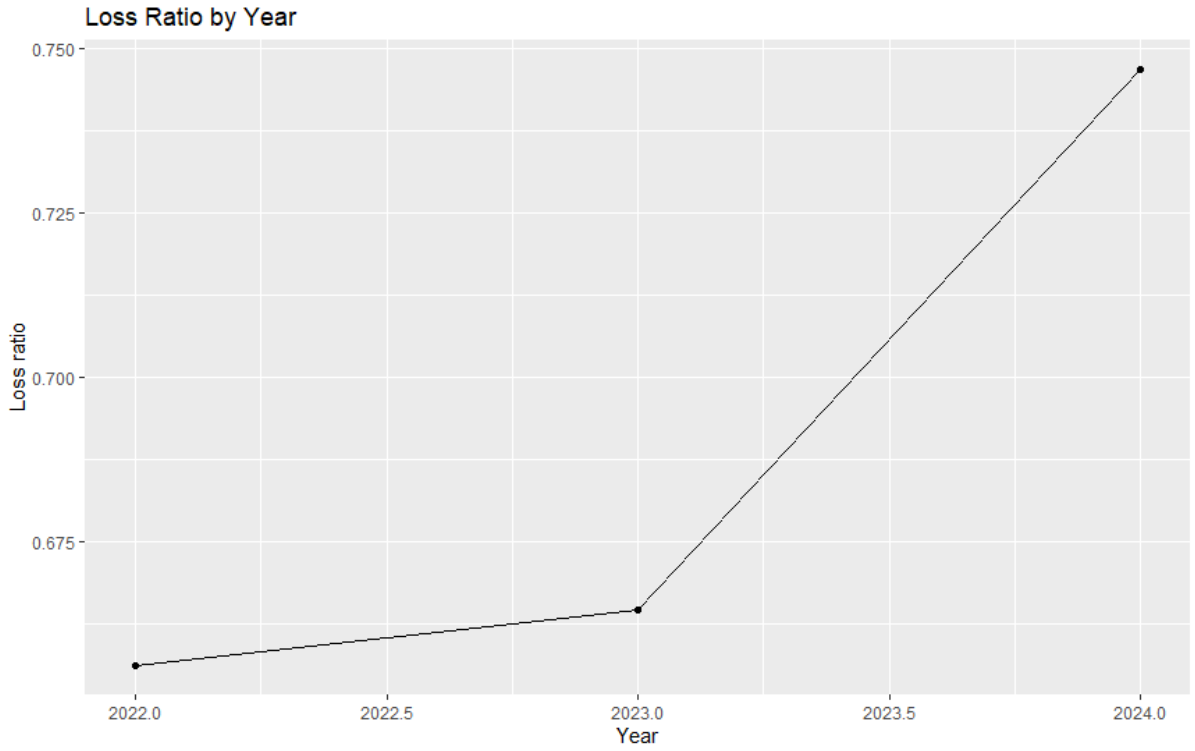


Figure 2. Annual incurred-loss ratio, 2022–2024.

Differences between policy types were more pronounced than the overall annual movement. COMP_N had the highest claim frequency (1.046) and loss ratio (93.9%), as well as the highest pure premium at approximately €787 per unit of exposure. By comparison, CC had a frequency of 0.221 and a loss ratio of 63.4%. These are descriptive comparisons, not evidence that policy type alone causes the differences; risk mix and exposure characteristics may also vary between groups.

Policy type	Claim frequency	Pure premium (€ / exposure)	Loss ratio
CC	0.221	218.83	63.4%
COMP_E	0.410	405.40	73.8%
COMP_N	1.046	786.66	93.9%
TPG	0.207	227.72	69.2%
TP	0.133	145.54	54.3%

Table 2. Selected policy-type risk measures.

4. Claim frequency modelling

Claim counts were modelled using a Poisson regression with log exposure included as an offset. The Poisson model provides a useful starting point, but its Pearson dispersion statistic was approximately 1.74, indicating more variation in claim counts than the model accounts for. The observed zero-claim proportion was 85.7%, compared with 81.7% under the fitted Poisson model, providing further evidence that the basic specification did not capture the count distribution fully.

A Negative Binomial model was therefore fitted to allow for overdispersion. The final model, which groups infrequent vehicle brands, had an AIC of approximately 366,559 and a Pearson dispersion statistic of 1.18. This was a substantial improvement over the Poisson benchmark, although the remaining dispersion and tail differences indicate that the fit is not perfect.

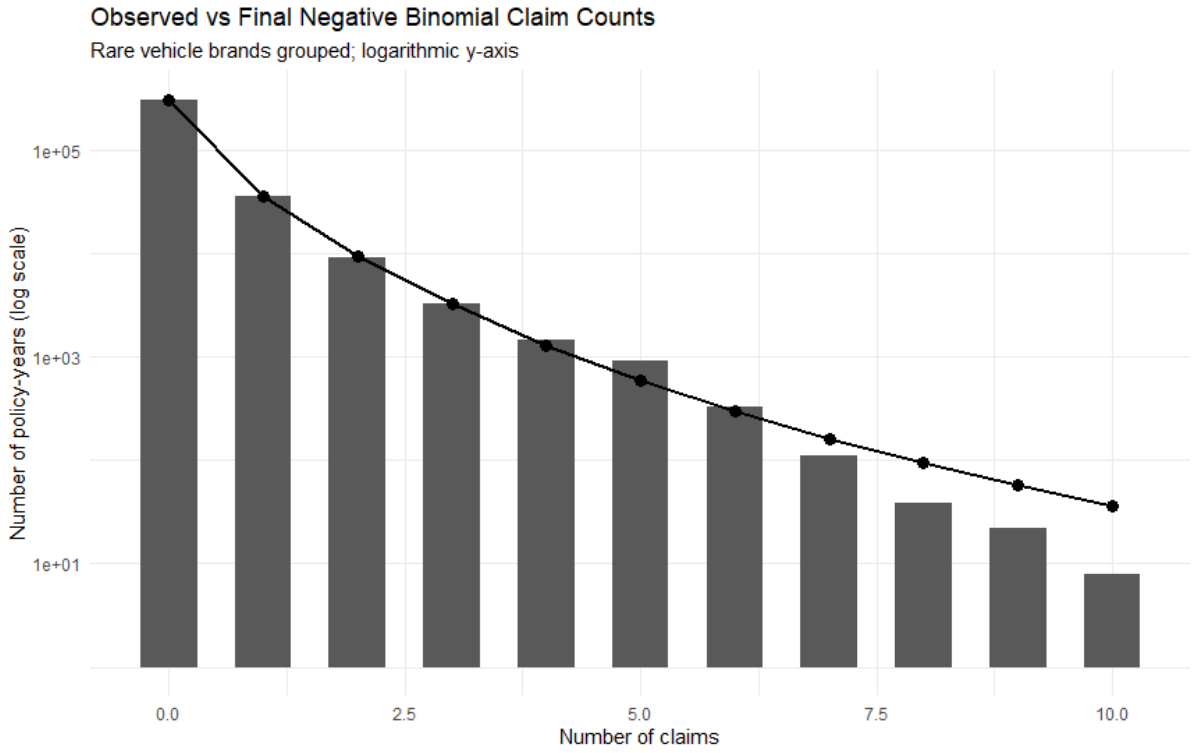


Figure 3. Observed and fitted claim-count distributions under the final Negative Binomial model (logarithmic vertical axis).

Out-of-sample validation used data from 2022–2023 for training and 2024 for testing. The model predicted approximately 37,938 claims, compared with 39,276 observed, an underprediction of 3.41%. This is a reasonably close aggregate prediction, but it shows that the model did not fully capture the level of 2024 experience. One holdout year is also too limited to establish that the model will perform equally well in other periods.

5. Claim severity and extreme-value modelling

5.1 Severity distribution

The positive-severity sample was strongly right-skewed. The mean policy-year average severity was €1,019.48, compared with a median of €643.83, and the standard deviation was approximately €3,182. The maximum observed value was €244,091. The large gap between the mean and median, together with the high maximum, shows that a small number of large observations exert a strong influence on the severity distribution.

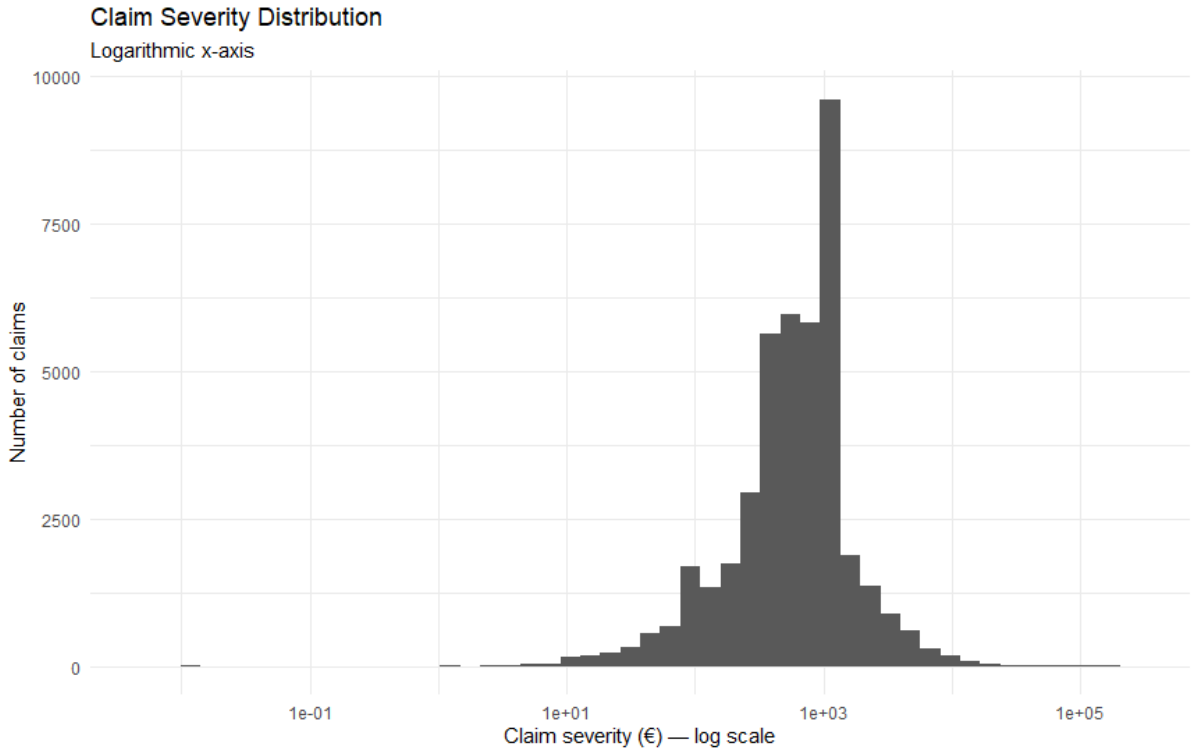


Figure 4. Distribution of positive incurred claim severity (logarithmic x-axis).

Gamma and Lognormal distributions were fitted as candidate severity models. The Lognormal fit was clearly better by the information criteria: its AIC was approximately 79,908, compared with 86,100 for Gamma, while its BIC was approximately 79,925 compared with 86,117. The Lognormal is therefore the stronger overall parametric benchmark in this analysis. However, an improved overall fit does not guarantee that the most extreme claims are represented accurately.

Statistic	Empirical	Gamma fit	Lognormal fit
99th percentile	€7,087	€4,807	€7,551
99.5th percentile	€10,543	€5,542	€9,990
99.9th percentile	€28,810	€7,250	€17,794

Statistic	Empirical	Gamma fit	Lognormal fit
99.95th percentile	€59,493	€7,986	€22,279
99.99th percentile	€139,174	€9,697	€36,034

Table 3. Empirical and fitted upper severity quantiles.

The comparison shows why the tail needs separate attention. The Lognormal is close to the empirical 99th and 99.5th percentiles, but it increasingly underestimates severity further into the tail. At the 99.99th percentile, the empirical value is about €139,174, compared with approximately €36,034 under the Lognormal fit. The Gamma fit underestimates the upper quantiles even more strongly.

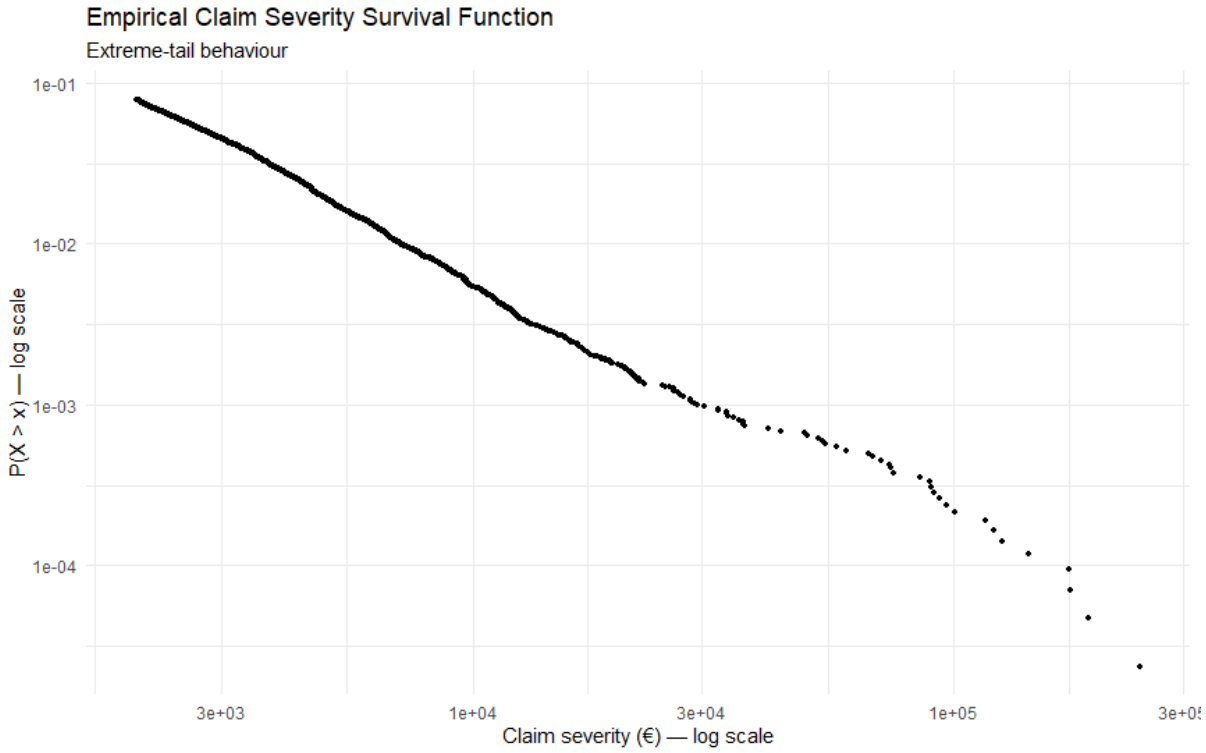


Figure 5. Empirical severity survival function on logarithmic axes.

5.2 Extreme Value Theory

Extreme Value Theory was used to model observations above a high severity threshold with a Generalized Pareto Distribution (GPD). A threshold of €10,000 was selected as a practical compromise among the tested thresholds; it was not uniquely optimal. This left 230 exceedances, or approximately 0.544% of the severity sample. The estimated shape parameter

was 0.825, and the scale parameter was approximately €5,286.

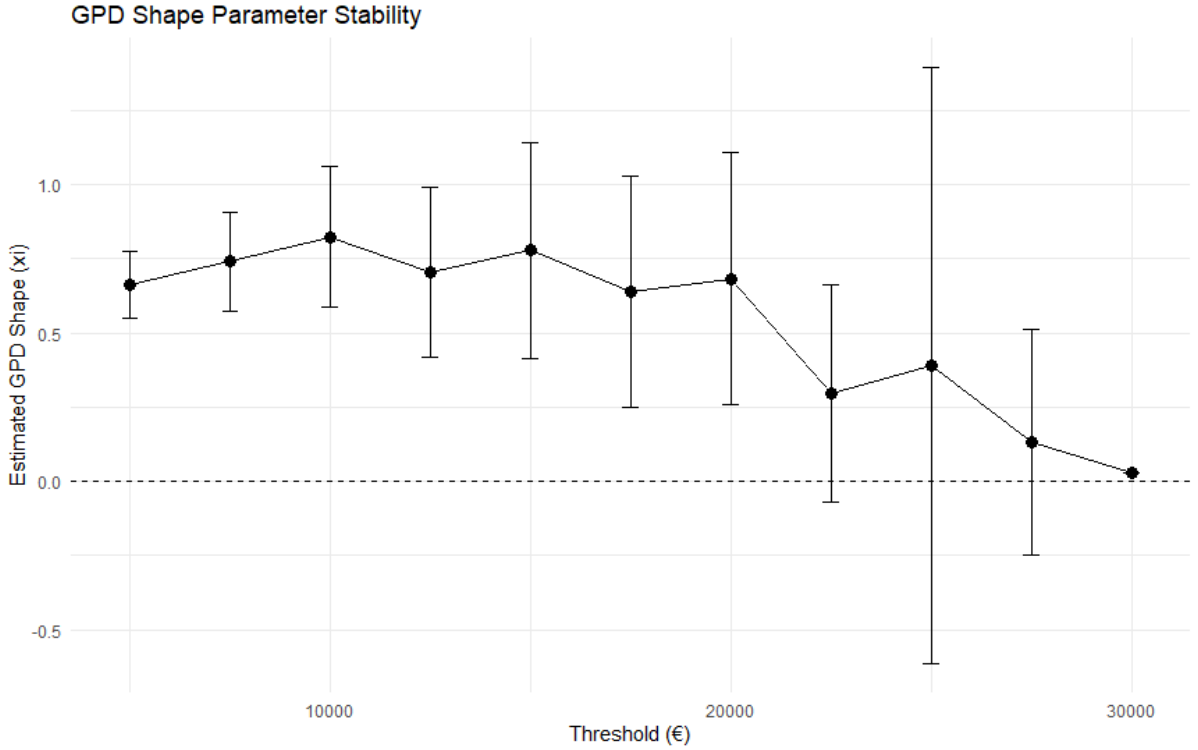


Figure 6. GPD shape parameter estimates across candidate thresholds.

The estimated shape parameter indicates a heavy tail. Its 95% Wald interval was approximately 0.589 to 1.060 and therefore included values above one. This matters because, under the fitted GPD, values of the shape parameter above 0.5 imply an infinite theoretical variance, while values at or above one also implies an infinite theoretical mean. These are properties of the fitted tail model rather than claims about the true portfolio, but they underline the uncertainty involved in extrapolating far beyond the observed data. EVT offers a way to explore the tail, not a guarantee of precise extreme-loss estimates.

6. Aggregate loss simulation

The final stage combined claim frequency and severity assumptions in a Monte Carlo simulation, using 25,000 simulations to estimate the aggregate-loss distribution. VaR is the loss quantile at a chosen confidence level; TVaR is the expected loss in the corresponding upper tail. Results were produced for a Lognormal severity benchmark and a model that replaces the fitted

Lognormal tail with the GPD-based EVT tail.

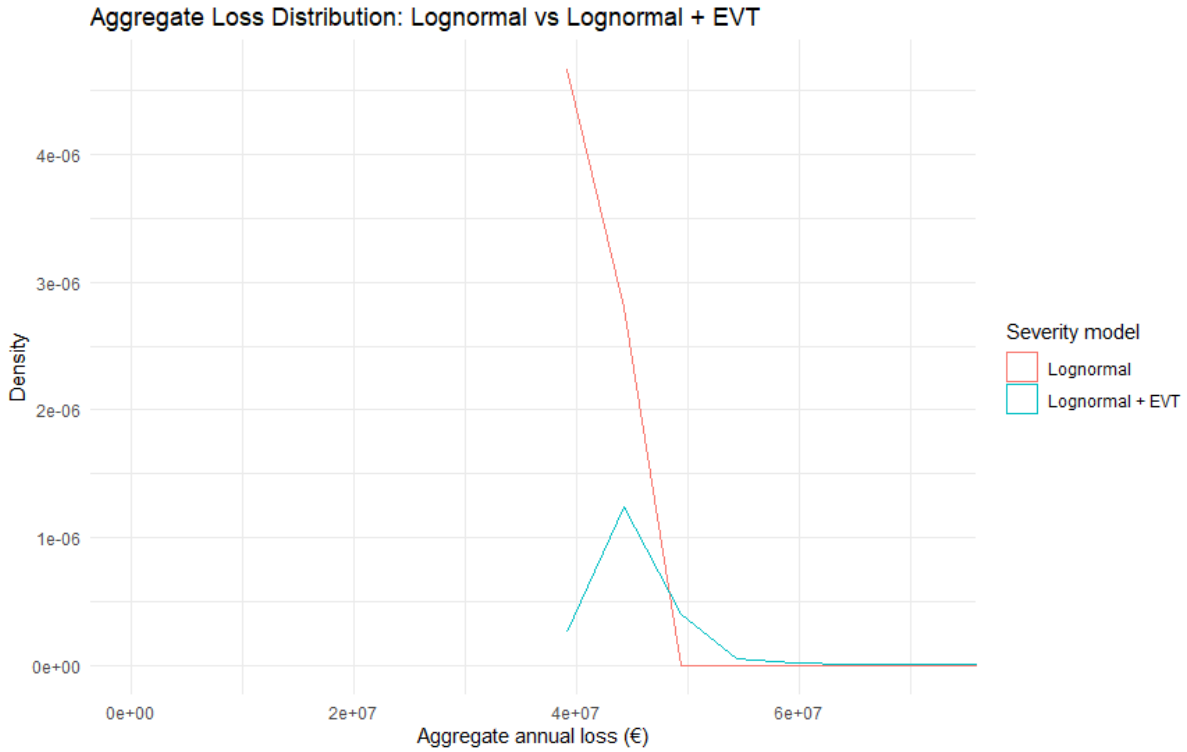


Figure 7. Simulated aggregate-loss distributions for the Lognormal and Lognormal + EVT approaches.

The figure shows a broad distributional comparison, but its horizontal axis is truncated at the 99.5th percentile. It should therefore not be used on its own to judge the most extreme losses; the risk-measure table below is more informative about the upper tail.

Confidence level	Severity approach	VaR (€m)	TVaR (€m)
99%	Lognormal	41.83	41.98
99%	Lognormal + EVT	72.17	148.48
99.5%	Lognormal	41.94	42.09
99.5%	Lognormal + EVT	94.98	215.25
99.9%	Lognormal	42.18	42.30
99.9%	Lognormal + EVT	245.69	546.75

Table 4. Simulated aggregate VaR and TVaR under alternative severity-tail assumptions.

Adding the EVT tail increased simulated risk measures sharply, particularly at 99.9% confidence. The Lognormal approach produced a 99.9% VaR of approximately €42.18 million and TVaR of €42.30 million, while the Lognormal + EVT approach produced approximately €245.69 million and €546.75 million, respectively. This difference demonstrates how sensitive aggregate-risk estimates can be to assumptions about rare, high-cost claims. Given the small number of exceedances and the uncertainty in the fitted GPD shape parameter, the EVT results should be interpreted as a stress-sensitive scenario rather than a reliable single-point estimate of required capital.

The separate simulation-size check also showed that estimates at the most extreme confidence level, especially TVaR, were sensitive to the number of simulations. Increasing the simulation count reduces Monte Carlo sampling error, but it does not remove uncertainty in the underlying frequency model, tail parameters or data construction.

7. Discussion and limitations

Three findings stand out. First, the portfolio grew substantially between 2022 and 2024 and the incurred-loss ratio increased, so absolute loss growth should be considered alongside exposure and premium growth. Second, the Negative Binomial model was better suited than Poisson to the variability in claim counts, although its 2024 forecast remained slightly low. Third, the Lognormal model described the central severity distribution better than Gamma, but it did not reproduce the most extreme observations; the EVT alternative exposed the potential scale of this tail risk.

The most important limitation concerns the severity data. The value used as “claim severity” is calculated as total incurred loss divided by total claim count for each policy-year record. It is therefore an average per claim count within a policy-year, not an individual claim transaction. Treating these averages as if they were individual claim amounts in an aggregate-loss simulation changes the distribution being modelled and may materially distort the tail. The simulated portfolio risk figures should consequently be regarded as exploratory, not actuarially validated estimates.

There are further limitations. The frequency model underpredicts 2024 claims and has only one holdout year available for validation. The GPD fit is based on 230 exceedances and the confidence interval for its shape parameter is wide enough to include very different tail behaviours. The highest simulated risk measures are also Monte Carlo-sensitive. Finally, records with claims but zero incurred loss and the isolated zero-exposure record with a claim require clarification before these data are used for operational pricing or capital decisions.

A stronger follow-up would use individual claim-level severity records, validate the frequency and severity components on later years or a separate portfolio, and test alternative EVT thresholds and tail specifications. Parameter uncertainty should also be propagated into aggregated risk measures. These steps would help distinguish true portfolio tail risk from uncertainty created by the modelling assumptions.

8. Conclusion

This project developed an end-to-end workflow for analyzing motor insurance risk, from cleaning and SQL-based portfolio summaries to regression modelling, severity fitting, EVT and Monte Carlo aggregation. The Negative Binomial model captured frequency variation better than Poisson, and the Lognormal distribution fitted overall severity better than Gamma. However, both severity fits understated the most extreme observations, and incorporating an EVT tail produced much higher simulated VaR and TVaR.

The main conclusion is that tail assumptions can dominate the estimated aggregate risk at high confidence levels. At the same time, the policy-year average severity data and uncertainty in the EVT fit prevent the simulated figures from being treated as definitive. The analysis is most useful as a demonstration of the modelling process and the importance of tail risk, with claim-level data and more extensive validation needed before the results could support real-world pricing or capital decisions.

References

Espinosa, P., Lledó, J. and Atance, D. (2026) 'A detailed dataset of motor insurance policies with coverage-specific financial information', *Data in Brief*, 66, 112883. Available at: https://doi.org/10.1016/j.dib.2026.112883.

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